

eFinder Hub

The Nexus eFinder has been designed to primarily interface to a Nexus DSC Pro. The Live mode was included for those with 2nd or more scopes without encoders.

A lot of people are just using Live Mode. Ideally SkySafari will one day get connection capability for digital finders, but the current arrangement works, given the Nexus eFinder has no display or controls of its own. There are many who do not want to use SkySafari, or indeed any phone or tablet. This is usually to protect their dark adaptation, but for other reasons too.

For these observers and perhaps others, I propose a new 'Hub' device. It would comprise a Raspberry Pi (or course!), a red filtered OLED display (very similar to the Nexus DSC display) and a simple 5-way navigation switch. This display and switch will be familiar to eFinder Lite and ScopeDog users. Below is a photo of my prototype (no case yet).

Starting up

1. The Hub would be powered from any usb source. (USB C socket on the hub)
2. Power up the Hub and wait for the display to say "plug in eFinder"
3. Use the (supplied) USB A to right angle USB C to connect Nexus eFinder to Hub
4. Display reports connection and eFinder boot progress. Note it uses 'Nexus Mode'
5. Displays the 'Home' screen.
6. Hub will either broadcast a Hotspot wifi or connect to a preconfigured infrastructure network (wifi or ethernet)

Stand alone operation

7. Navigate to Offset screen to perform Offset calibration
8. Optional - navigate to Solve Assist screen to adjust exposure and check solving OK
9. Optional - navigate to Display Brightness adjust (value is saved for next time)
10. Optional - navigate to Date adjust. Set year and month to enable precession to be applied. (Value saved)
11. Go to Home Screen.
12. Press OK for a single solve.
13. Use up or down button to flip to Auto mode, where it will continuously solve as fast as possible.

Connected to infrastructure wifi network.

14. Set your Long & Lat in the hub.config file
15. The Hub will get UTC time from the internet.
16. The AltAz screen to the left of Home, will now be available.

Connected to SkySafari

17. After step 6 above. Connect SkySafari over WIFI (same settings as for Live mode)
18. SkySafari will set & override location, date and time in the Hub
19. The AltAz screen to the left of Home, will now be available.
20. Offset can now be done from the appropriate Hub screen, or the Align button on the SkySafari screen.
21. Navigate to the AltAz screen (left of Home) & select auto mode
22. A target can be selected on SkySafari and Goto commanded.
23. The Hub screen will change to show direction and distance to target.
24. Push to target but remember the Hub display is only updated when the scope is not moving.

Use Optional usb GPS dongle

25. Plug in dongle before powering up the Hub.
26. Start up as described above.
27. During boot, the process will pause while it acquires a GPS fix. This will be displayed on the screen. This fix supplies accurate location and time data and thus enables the AltAz screen

Photos of prototype



Arrangement of screens and sample images

